

	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 09-07-2024	
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
June 2024



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1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed		Observations		
ms^{-1}		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	392	160	34
	4.2 < ws < 7.6	173	297	189
	ws > 7.6	14	125	1094
Number of points 2478 ; PC=72.0%; EBD=1.9%; POD1=67.7%; POD2=51.0%; POD3=83.1%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	546	36	0
	4.2 < ws < 7.6	33	522	55
	ws > 7.6	0	24	1262
Number of points 2478 ; PC=94.0%; EBD=0.0%; POD1=94.3%; POD2=89.7%; POD3=95.8%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed	Observations			
ms^{-1}	ws < 12.0	12.0 < ws < 18.0	ws > 18.0	
Model data	ws < 12.0	1944	42	0
	12.0 < ws < 18.0	17	452	5
	ws > 18.0	0	8	10
Number of points 2478 ; PC=97.1%; EBD=0.0%; POD1=99.1%; POD2=90.0%; POD3=66.7%				

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

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wind direction degree		North	East	South	Observations	West
Model data	North	1717	102	0		0
	East	16	70	7		0
	South	11	4	78		0
	West	85	2	0		10
Number of points 2102 ; PC=89.2%; EBD=0.6%; POD1=93.9%; POD2=39.3%; POD3=91.8%; POD4=100.0%						

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	1819	16	0		0
	East	10	157	7		0
	South	0	5	78		0
	West	0	0	0		10
Number of points 2102 ; PC=98.2%; EBD=0.0%; POD1=99.5%; POD2=88.2%; POD3=91.8%; POD4=100.0%						

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	695	155	0
	7.1 < rh < 15.3	353	731	140
	rh > 15.3	2	113	290
Number of points 2479 ; PC=69.2%; EBD=0.1%; POD1=66.2%; POD2=73.2%; POD3=67.4%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1011	29	0
	7.1 < rh < 15.3	39	959	10
	rh > 15.3	0	11	420
Number of points 2479 ; PC=96.4%; EBD=0.0%; POD1=96.3%; POD2=96.0%; POD3=97.7%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	2479	0	0
	50.0 < rh < 70.0	0	0	0
	rh > 70.0	0	0	0
Number of points 2479 ; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=NaN%; POD3=NaN%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1148	224	0
	1.9 < pwv < 3.5	107	463	75
	pwv > 3.5	0	0	392
Number of points 2409 ; PC=83.1%; EBD=0.0%; POD1=91.5%; POD2=67.4%; POD3=83.9%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1240	16	0
	1.9 < pwv < 3.5	15	670	4
	pwv > 3.5	0	1	463
Number of points 2409 ; PC=98.5%; EBD=0.0%; POD1=98.8%; POD2=97.5%; POD3=99.1%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

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total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	318	125	17
	0.6 < see < 0.8	196	139	102
	see > 0.8	6	44	113
Number of points 1060 ; PC=53.8%; EBD=2.2%; POD1=61.2%; POD2=45.1%; POD3=48.7%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	512	48	0
	0.6 < see < 0.8	49	231	13
	see > 0.8	0	23	279
Number of points 1060 ; PC=88.5%; EBD=0.0%; POD1=91.3%; POD2=76.5%; POD3=95.5%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	517	0	12
	0.6 < see < 0.8	0	292	12
	see > 0.8	3	16	220
Number of points 1060 ; PC=97.1%; EBD=1.4%; POD1=99.4%; POD2=94.8%; POD3=90.2%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	561	0	0
	0.6 < see < 0.8	0	302	0
	see > 0.8	0	0	292
Number of points 1060 ; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=100.0%; POD3=100.0%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	129	65	22
	$3.4 < \tau < 5.8$	57	179	93
	$\tau > 5.8$	6	84	282
Number of points 917 ; PC=64.3%; EBD=3.1%; POD1=67.2%; POD2=54.6%; POD3=71.0%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	177	6	1
	$3.4 < \tau < 5.8$	15	295	35
	$\tau > 5.8$	0	27	361
Number of points 917 ; PC=90.8%; EBD=0.1%; POD1=92.2%; POD2=89.9%; POD3=90.9%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	186	0	82
	$3.4 < \tau < 5.8$	0	296	82
	$\tau > 5.8$	6	32	315
Number of points 917 ; PC=86.9%; EBD=9.6%; POD1=96.9%; POD2=90.2%; POD3=65.8%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	192	0	4
	$3.4 < \tau < 5.8$	0	323	4
	$\tau > 5.8$	0	5	393
Number of points 917 ; PC=99.0%; EBD=0.4%; POD1=100.0%; POD2=98.5%; POD3=98.0%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	93	125
	0.6 < glf < 0.7	83	124
	glf > 0.7	40	98
Number of points 936 ; PC=35.6%; EBD=10.1%; POD1=43.1%; POD2=35.7%; POD3=31.1%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	174	33
	0.6 < glf < 0.7	42	259
	glf > 0.7	0	55
Number of points 936 ; PC=80.8%; EBD=0.1%; POD1=80.6%; POD2=74.6%; POD3=86.6%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	172	18
	0.6 < glf < 0.7	4	323
	glf > 0.7	40	6
Number of points 936 ; PC=86.2%; EBD=10.8%; POD1=79.6%; POD2=93.1%; POD3=72.4%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	216	1
	0.6 < glf < 0.7	0	345
	glf > 0.7	0	1
Number of points 936 ; PC=99.7%; EBD=0.1%; POD1=100.0%; POD2=99.4%; POD3=99.5%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	721	157
	4.6 < ws < 7.4	271	150
	ws > 7.4	9	36
Number of points 2546 ; PC=76.7%; EBD=0.5%; POD1=72.0%; POD2=43.7%; POD3=89.9%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	934	39
	4.6 < ws < 7.4	67	274
	ws > 7.4	0	30
Number of points 2546 ; PC=94.1%; EBD=0.0%; POD1=93.3%; POD2=79.9%; POD3=98.8%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	1840	53
	12.0 < ws < 18.0	21	451
	ws > 18.0	0	8
Number of points 2546 ; PC=95.6%; EBD=0.0%; POD1=98.9%; POD2=88.1%; POD3=82.7%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	1493	64	0		10
	East	15	45	16		0
	South	10	5	48		20
	West	166	7	10		149
Number of points 2058 ; PC=84.3%; EBD=0.8%; POD1=88.7%; POD2=37.2%; POD3=64.9%; POD4=83.2%						

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	1650	7	0		7
	East	15	111	4		0
	South	0	1	67		0
	West	19	2	3		172
Number of points 2058 ; PC=97.2%; EBD=0.1%; POD1=98.0%; POD2=91.7%; POD3=90.5%; POD4=96.1%						

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	762	272	0
	8.9 < rh < 17.9	236	762	219
	rh > 17.9	0	59	236
Number of points 2546 ; PC=69.1%; EBD=0.0%; POD1=76.4%; POD2=69.7%; POD3=51.9%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	949	38	0
	8.9 < rh < 17.9	49	1047	14
	rh > 17.9	0	8	441
Number of points 2546 ; PC=95.7%; EBD=0.0%; POD1=95.1%; POD2=95.8%; POD3=96.9%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	2546	0	0
	50.0 < rh < 70.0	0	0	0
	rh > 70.0	0	0	0
Number of points 2546 ; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=NaN%; POD3=NaN%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	1115	160	0
	2.1 < pwv < 3.7	126	576	182
	pwv > 3.7	0	26	361
Number of points 2546 ; PC=80.6%; EBD=0.0%; POD1=89.8%; POD2=75.6%; POD3=66.5%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	1218	20	0
	2.1 < pwv < 3.7	23	733	8
	pwv > 3.7	0	9	535
Number of points 2546 ; PC=97.6%; EBD=0.0%; POD1=98.1%; POD2=96.2%; POD3=98.5%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}		Observations		
		$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	368	162	18
	$4.2 < ws < 7.6$	151	268	181
	$ws > 7.6$	61	104	1099
Number of points 2412 ; PC=71.9%; EBD=3.3%; POD1=63.4%; POD2=50.2%; POD3=84.7%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree		Observations			
		North	East	South	West
Model data	North	1670	84	7	6
	East	39	66	4	0
	South	18	14	72	0
	West	49	0	2	4

Number of points 2035 ; PC=89.0%; EBD=1.2%; POD1=94.0%; POD2=40.2%; POD3=84.7%; POD4=40.0%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	599	210	0
	7.1 < rh < 15.3	425	651	104
	rh > 15.3	18	79	326

Number of points 2412 ; PC=65.3%; EBD=0.7%; POD1=57.5%; POD2=69.3%; POD3=75.8%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1048	168	0
	1.9 < pwv < 3.5	140	518	166
	pwv > 3.5	0	1	301

Number of points 2342 ; PC=79.7%; EBD=0.0%; POD1=88.2%; POD2=75.4%; POD3=64.5%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	392	142	15
	0.6 < see < 0.8	221	177	135
	see > 0.8	59	96	255
Number of points 1492 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	626	0	10
	0.6 < see < 0.8	0	362	10
	see > 0.8	46	53	395
Number of points 1492 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	369	176	56
	$3.4 < \tau < 5.8$	75	203	95
	$\tau > 5.8$	3	61	274
Number of points 1312 ; PC=64.5%; EBD=4.5%; POD1=82.6%; POD2=46.1%; POD3=64.5%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	440	17	94
	$3.4 < \tau < 5.8$	4	408	94
	$\tau > 5.8$	3	15	331
Number of points 1312 ; PC=89.9%; EBD=7.4%; POD1=98.4%; POD2=92.7%; POD3=63.8%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	131	115
	0.6 < glf < 0.7	205	303
	glf > 0.7	33	91
Number of points 1377 ; PC=44.1%; EBD=8.4%; POD1=35.5%; POD2=59.5%; POD3=34.7%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	324	20
	0.6 < glf < 0.7	12	484
	glf > 0.7	33	5
Number of points 1377 ; PC=87.7%; EBD=9.6%; POD1=87.8%; POD2=95.1%; POD3=67.6%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	719	163	1
	4.6 < ws < 7.4	278	159	111
	ws > 7.4	4	21	1090
Number of points 2546 ; PC=77.3%; EBD=0.2%; POD1=71.8%; POD2=46.4%; POD3=90.7%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	1554	70	0
	East	7	30	2
	South	3	21	60
	West	120	0	12

Number of points 2058 ; PC=87.8%; EBD=0.1%; POD1=92.3%; POD2=24.8%; POD3=81.1%; POD4=91.1%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	684	237
	8.9 < rh < 17.9	314	805
	rh > 17.9	0	51

Number of points 2546 ; PC=67.6%; EBD=0.7%; POD1=68.5%; POD2=73.7%; POD3=51.2%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	1096	157
	2.1 < pwv < 3.7	145	560
	pwv > 3.7	0	45

Number of points 2546 ; PC=78.6%; EBD=0.0%; POD1=88.3%; POD2=73.5%; POD3=63.5%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed <i>ms</i> ⁻¹		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	385	202	22
	4.2 < ws < 7.6	160	177	132
	ws > 7.6	35	141	1091
Number of points 2345 ; PC=70.5%; EBD=2.4%; POD1=66.4%; POD2=34.0%; POD3=87.6%				

Table 47: night3 - Contingency table for wind speed (*ms*⁻¹) in standard configuration

FATE - Monthly Report

wind direction degree		Observations			
		North	East	South	West
Model data	North	1597	53	6	1
	East	34	51	7	0
	South	66	23	72	2
	West	48	1	0	7

Number of points 1968 ; PC=87.8%; EBD=3.7%; POD1=91.5%; POD2=39.8%; POD3=84.7%; POD4=70.0%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	538	230	0
	7.1 < rh < 15.3	475	602	146
	rh > 15.3	29	49	276

Number of points 2345 ; PC=60.4%; EBD=1.2%; POD1=51.6%; POD2=68.3%; POD3=65.4%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1012	219	0
	1.9 < pwv < 3.5	109	468	254
	pwv > 3.5	0	0	213

Number of points 2275 ; PC=74.4%; EBD=0.0%; POD1=90.3%; POD2=68.1%; POD3=45.6%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	367	128	20
	0.6 < see < 0.8	240	178	107
	see > 0.8	65	109	260
Number of points 1474 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	611	0	11
	0.6 < see < 0.8	0	390	11
	see > 0.8	61	25	376
Number of points 1474 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	317	91	5
	$3.4 < \tau < 5.8$	103	289	139
	$\tau > 5.8$	0	60	281
Number of points 1285 ; PC=69.0%; EBD=0.4%; POD1=75.5%; POD2=65.7%; POD3=66.1%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	414	5	85
	$3.4 < \tau < 5.8$	6	398	85
	$\tau > 5.8$	0	37	340
Number of points 1285 ; PC=89.6%; EBD=6.6%; POD1=98.6%; POD2=90.5%; POD3=66.7%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	122	103
	0.6 < glf < 0.7	222	300
	glf > 0.7	18	95
Number of points 1350 ; PC=44.3%; EBD=5.9%; POD1=33.7%; POD2=60.2%; POD3=35.9%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	333	9
	0.6 < glf < 0.7	11	486
	glf > 0.7	18	3
Number of points 1350 ; PC=91.9%; EBD=6.4%; POD1=92.0%; POD2=97.6%; POD3=75.3%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	594	142	1
	4.6 < ws < 7.4	404	147	126
	ws > 7.4	3	54	1075
Number of points 2546 ; PC=71.3%; EBD=0.2%; POD1=59.3%; POD2=42.9%; POD3=89.4%				

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind direction degree	Observations			
	North	East	South	West
Model data	North	1559	77	0
	East	0	36	9
	South	19	8	52
	West	106	0	13

Number of points 2058 ; PC=86.6%; EBD=0.9%; POD1=92.6%; POD2=29.8%; POD3=70.3%; POD4=76.0%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	652	253
	8.9 < rh < 17.9	298	723
	rh > 17.9	48	117

Number of points 2546 ; PC=61.8%; EBD=1.9%; POD1=65.3%; POD2=66.1%; POD3=43.7%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	1115	248
	2.1 < pwv < 3.7	126	484
	pwv > 3.7	0	30

Number of points 2546 ; PC=76.0%; EBD=0.0%; POD1=89.8%; POD2=63.5%; POD3=61.9%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration